Do Now · retrieval

Run-length and dictionary coding

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Define lossless compression.

Q2

What is a run in RLE?

Q3

What is a compression overhead?

Q4

For array [2,3,4,5], double values at indexes 1 and 3. State final array and sum.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Understand — Run-length encoding replaces each maximal run of the same symbol with a value and count; it works best for long consecutive repetitions.

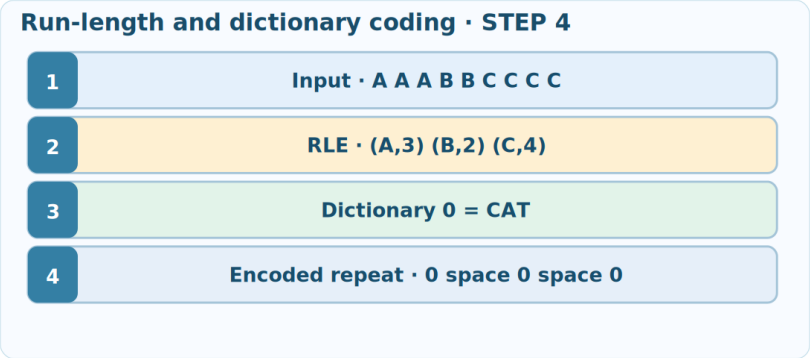
Core idea 2

Apply — Dictionary coding replaces repeated strings with dictionary references; the decoder needs the same dictionary, transmitted or reconstructed according to the scheme.

Core idea 3

Evaluate — Overhead matters. Alternating characters can make naive RLE larger; long repeated phrases can be valuable for dictionary coding.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

RLE — Encode repeated runs with counts.

run — Maximal consecutive equal symbols.

dictionary coding — Represent repeated strings by references.

dictionary — Mapping of codes to values/phrases.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

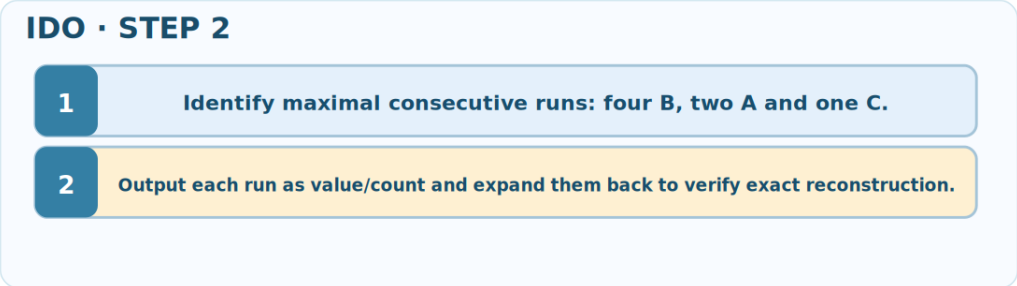
Encode BBBBAAC as symbol/count pairs and verify decoding.

Follow the worked method

1. Predict first — Identify maximal consecutive runs: four B, two A and one C.

2. Model and check — Output each run as value/count and expand them back to verify exact reconstruction.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first

Use dictionary reference 1 for each repeated word while preserving the space separator.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

RLE-encode AAAAABBCCCDDDD with counts. [4]

Question 2

With dictionary 0='HELLO',1='WORLD', encode 'HELLO WORLD HELLO' using references and separators. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain why RLE can increase size for alternating data and why dictionary coding may help repeated long substrings. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

RLE always shortens any input file.

Use key vocabulary from memory

1. Explain RLE in your own words.

2. Explain run in your own words.

3. Explain dictionary coding in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.